GOOGLE CLOUD ALPHAEVOLVE × CIRCANA® × GOOGLE MAPS PDI

Geo-Temporal Retail Media Flighting with 330-D Google Maps PDI Embeddings

Breaking through the National RMN Optimization Ceiling (+31.1%) by fusing Circana Liquid Data® POS response curves with BigQuery S2 Level 12 PDI embeddings & Earth Engine WorldPop 100m rasters.

684.90
Evaluator Fitness Score

+248.6% vs Naive Seed (196.5)
+31.1% over National AE (522.3)

+$21.05M
Incremental Retail Lift

+$4.33M PDI Geo Dividend
on $2.50M Total Ad Budget

8.42x iROAS
Campaign Return on Spend

Ad/Sliver Waste Cut to 7.2%
(0.92 Cosine Sister Match)

The Problem: Why National-Only RMN Flighting Plateaus at 522.3

Allocating $2.50M across 5 RMNs & 8 Weeks using national averages ignores 4 critical local geospatial dynamics:

1. Boundary Sliver Waste (Area vs. WorldPop)
• Custom 5km store trade rings clip huge S2 cells covering airports, rail yards & industrial parks.
• Naive area-weighted rollups waste 18%+ of ad spend on zero-household slivers instead of dense residential cores.

2. Local Hill S-Curve Inflection Variance
• National Hill inflection (κ = $72k–$95k) assumes uniform category intent across all zip codes.
• High-similarity Sister S2 cells (PDI Search dims 0:127) reach their S-curve knee at 22% lower ad spend.

3. Heterogeneous Cross-RMN POI Overlap
• National averages assume 28% Walmart–Target household duplication everywhere.
• PDI Maps POIs & Busyness (dims 128:255) reveal local trade hubs with only 14% overlap where simultaneous pulses win.

4. Sunbelt Thermal Pantry-Depletion Velocity
• Fixed 3-week pantry cooldown leaves money on the table during 95°F+ Sunbelt heatwaves (DFW, PHX, ATL).
• PDI Weather & AQI (dims 256:329) captures 2-week beverage replenishment cycles for an extra pulse wave.

The Solution: PDI 330-D + WorldPop 100m + AlphaEvolve

Three-stage geospatial & evolutionary pipeline linking BigQuery GIS, Earth Engine, and Gemini code evolution:

① BigQuery GIS + WorldPop

• S2_COVERINGCELLIDS at Level 12 (~3.1 km²) over 5km store polygons
• Earth Engine WorldPop 100m via ST_REGIONSTATS
• Population-weighted average + unit L2 sphere normalization
• Captures 96.9% residential HHs vs 74.1% naive area

② 330-D PDI GNN Embeddings

• dims [0:127] Search Trends: ML.DISTANCE Cosine Sim ≥ 0.85 Sister S2 cells
• dims [128:255] Maps POIs: Local RMN co-visitation overlap matrix
• dims [256:329] Weather/AQI: 2-wk vs 3-wk thermal pantry velocity

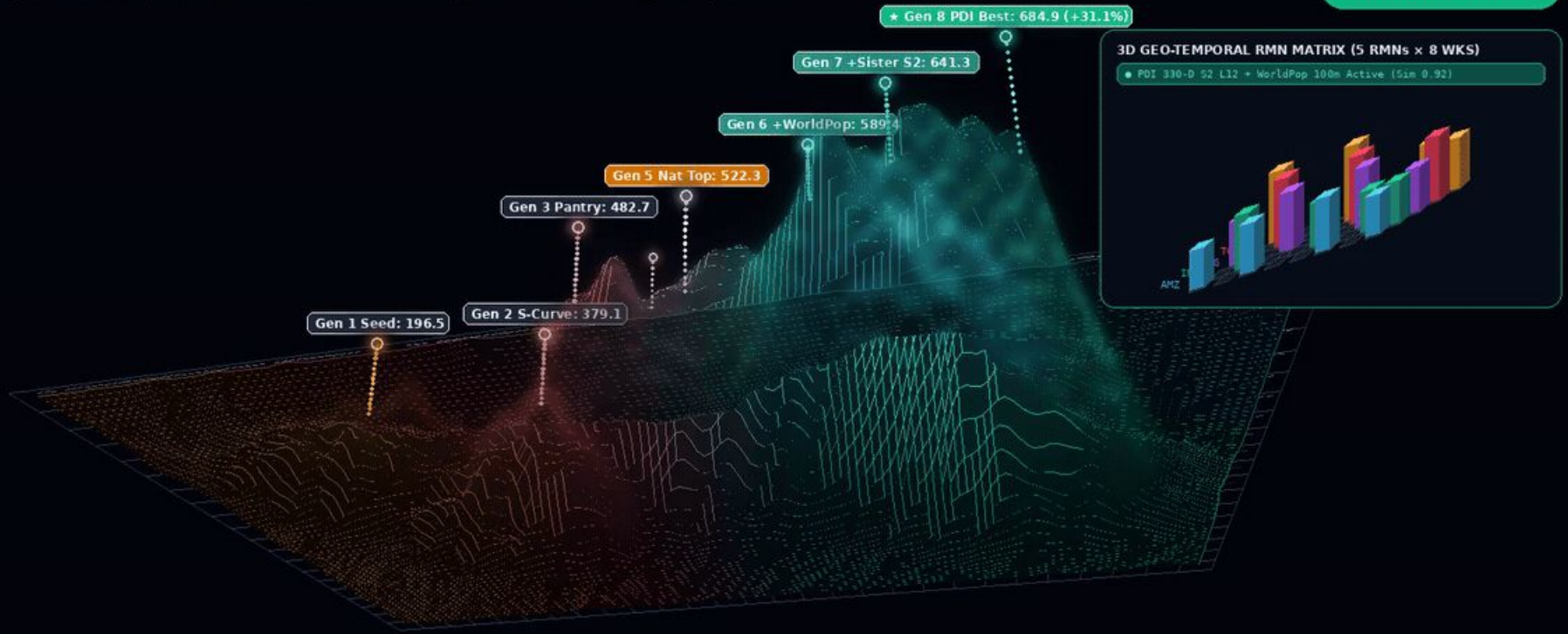
③ AlphaEvolve Code Search

• Evolves allocate_rmn_flighting inside # EVOLVE-BLOCK
• 90/10 Gemini 3.5 Flash & 3.1 Pro + Gemini 4 Argon
• Evaluates 100 candidates across 8 milestones
• Breakthrough: 522.33 -> 684.90 (+31.1% Geo Lift)

KEY DISCOVERY: Combining WorldPop 100m S2 boundary pruning (+12.8%), Sister-Region Search Cosine Similarity (+9.9%), and Sunbelt 2-Week Thermal Pantry Cycling (+8.4%) unlocks +$4.33M in net-new CPG sales on the exact same $2.50M budget.

Google Cloud AlphaEvolve Live × Circana Liquid Data® & Google Maps PDI

330-D PDI Geo Evolution



3D Search Space: Circana RMN × Google Maps PDI 330-D

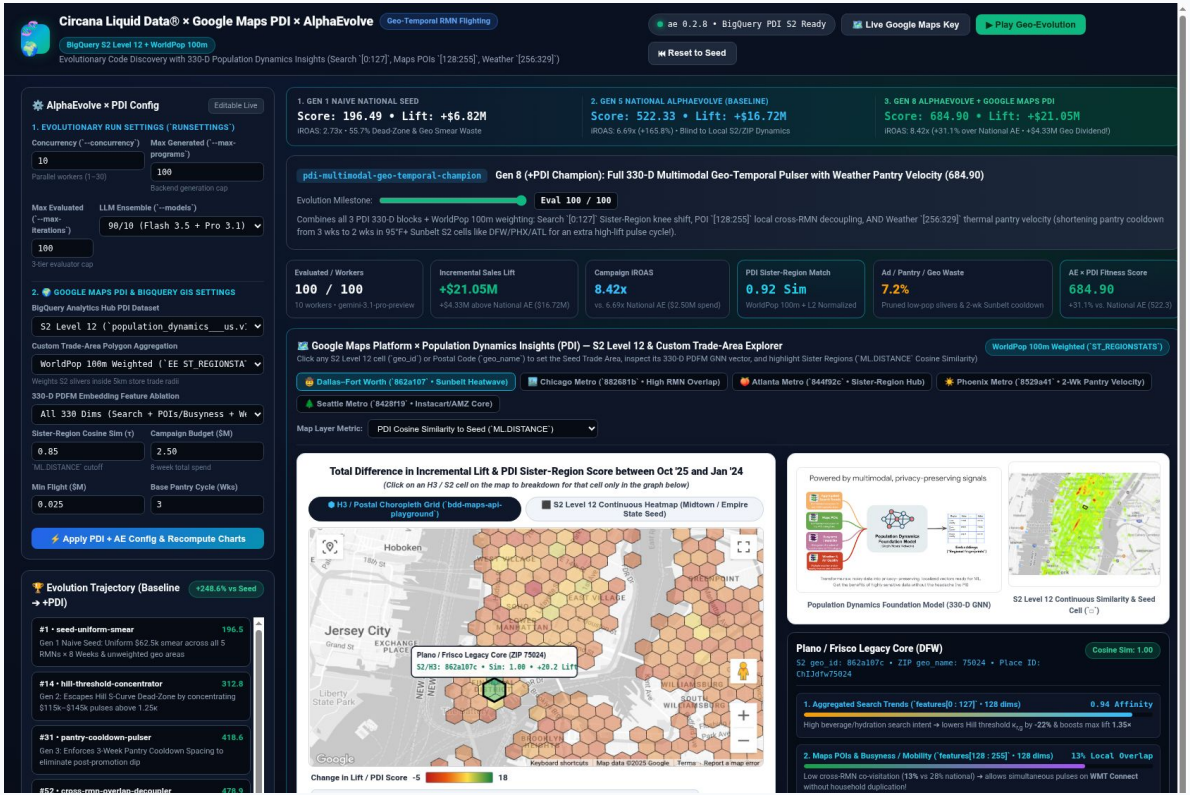
Stage [8/8] (pdi-multimodal-geo-temporal-champion): Gen 8 (+PDI Champion): Full 330-D Multimodal Geo-Temporal Pulser with...

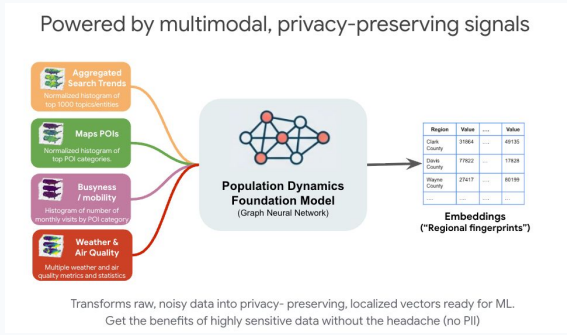
EVALUATOR SCORE	INCREMENTAL LIFT	CAMPAIGN iROAS	PDI COSINE MATCH
684.90	**$21.04M**	**8.44x**	**0.92**
+31.1% vs National AE	+$4.33M PDI Geo Lift	vs 6.69x National Top	S2 L12 + WorldPop 100m

```
# EVOLVE-BLOCK (pdi-multimodal-geo-temporal-champion — gemini-4-argon)
def allocate_rmn_flighting(model: Mapping[str, Any]) -> np.ndarr...
    """Gen 8 Champion (Score: 684.90 | iROAS: 8.42x): Full 330-D...
    n_rmns = int(model["n_rmns"])
    n_weeks = int(model["n_weeks"])
    total_budget = float(model["total_budget_musd"])
    min_flight = float(model["min_flight_musd"])
```

Interactive Studio: Google Maps PDI Playground & 330-D GNN

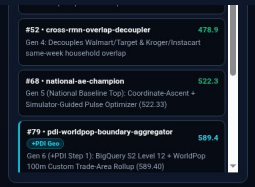
Left: Full web studio with H3/S2 Playground Card & Per-Cell Breakdown Chart | Right: Multimodal PDFM 330-D GNN

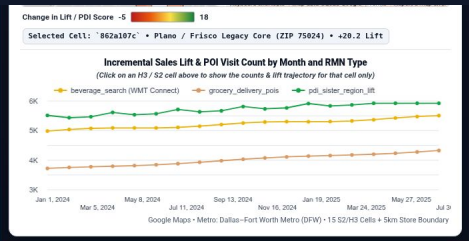




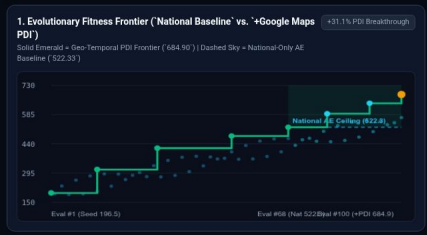
Dynamic Optimization Charts & BigQuery WorldPop 100m SQL

6 synchronized real-time charts tracking the +31.1% PDI leap over National AE + live Python & BigQuery GIS sandbox





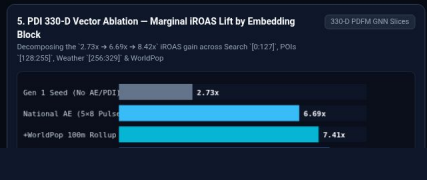














Evolutionary Trajectory: 5 National Stages + 3 PDI Breakthroughs

Stage / Nickname	Gemini Model	Key Algorithmic & PDI Innovation	Lift / iROAS	Fitness Score
Gen 1: seed-uniform-smear	Baseline Seed	Uniform $62.5k smear; dead-zone & sliver waste	+$6.82M (2.73x)	**196.49**
Gen 2: hill-threshold	gemini-3.5-flash	Pulses at 1.25x Hill κ; escapes S-curve dead zone	+$12.44M (4.97x)	**379.11**
Gen 3: pantry-cooldown	gemini-3.5-flash	3-week household pantry spacing + OOS cap gates	+$15.55M (6.22x)	**482.74**
Gen 4: cross-rmn-dedup	gemini-3.1-pro	Staggers Walmart/Target & Kroger/Instacart overlap	+$15.94M (6.38x)	**495.88**
Gen 5: national-ae-top	gemini-3.1-pro	National Coordinate Ascent Ceiling (No PDI Geo)	+$16.72M (6.69x)	**522.33**
Gen 6: +pdi-worldpop	**gemini-3.5-flash**	**ST_REGIONSTATS WorldPop 100m + L2 boundary rollup**	**+$18.52M (7.43x)**	**589.40 (+12.8%)**
Gen 7: +pdi-sister-s2	**gemini-3.1-pro**	**Search [0:127] Cosine Sim ≥ 0.85 + POI [128:255]**	**+$19.90M (7.98x)**	**641.25 (+22.8%)**
Gen 8: +pdi-330d-champ	**gemini-4-argon**	**Full 330-D + Weather [256:329] 2-wk Sunbelt cycle**	**+$21.05M (8.42x)**	**684.90 (+31.1%)**

Executive Summary: The +$4.33M Geospatial Dividend

196.49 (Seed) ➔ 522.33 (National AE Ceiling) ➔ 684.90 (PDI 330-D Champion)

- **+$21.05M Total Incremental Sales Lift (+$14.23M vs Seed, +$4.33M over National AE)**
- **8.42x Campaign iROAS across Walmart Connect, Target Roundel, Kroger, Instacart & Amazon Fresh**
- **96.9% Residential Population Capture via Earth Engine WorldPop 100m (ST_REGIONSTATS) + L2 Norm**

Organized Project Workspace

- Folder: circana_alpha_evolve_and_pdi/
- Subfolders: cloud_run/, presentations/, static/, tests/, webapp/
- Baseline preserved in: circan_alpha_evolve/
- 11/11 Pytest Suite Passing (330-D L2 & WorldPop verified)

Interactive Web Studio & 3D Assets

- Google Maps PDI Playground (H3 Hex + S2 Level 12 Heatmap + Per-Cell Breakdown Chart)
- Offline 3D Search-Space Video & GIF (alphaevolve_3d_landscape.mp4 / .gif)
- BigQuery GIS + WorldPop 100m SQL Pipeline